Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

University of Pisa, a.y. 2025-2026

Alessandro Gori*

Thesis for the Master's degree in Physics

Abstract

[To be continued...]

- Appendix on s-wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
- Set ComputerModern font in all CairoMakie figures

Contents

Draft: March 15, 2026

*a.gori23@studenti.unipi.it / nepero27178@github.com

List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Chapter 1

Antiferromagnetic instability

This chapter is devoted to an in-depth discussion of the antiferromagnetic (AF) long-range ordering in the EHM by the means of MFT. In the context of the 2D HM, discussed in App. **??**, AF ordering establishes as an unconventional metallic phase which is only stable at half-filling (as is discussed in Sec. **??**). The situation for the EHM is similar: the present analysis is carried out based on the argument that, given the symmetry structure of the non-local NN interaction, the fundamental features of the commensurate AF phase of the HM are effectively preserved, while the parameters are redefined by the interaction itself. The two most relevant effects we discuss in this chapter are the gap "complexification" and the bands renormalization and resymmetrization (both in Sec. 1.3), the latter being a general feature of the EHM common to all phases preserving (or, at most, weakening without destroying) translational invariance, discussed previously in Sec. **??**.

1.1 The AF phase and its symmetries

The AF phase is not uniquely defined; actually, there exists a variety of ways to establish a SDW phase in a spin-balanced square lattice, which means a system where both spin sectors are equally populated, but differently displaced. [ToDo: graphics of various SDW structures]

The AF phase hereby presented is a symmetry reduction of the normal phase, and is described by simultaneous spin rotational symmetry reduction and translational symmetry halving:

$$\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^z(1)$$

$$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \overset{\mathrm{SSB}}{\to} \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)}$$

We describe a SDW whose period is of two sites. As done for the normal phase, we allow theoretically

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$$

in order to capture the same anisotropy effects seen in Sec. **??**.

Recall Tabs. **??** and **??**, collecting the RWCs for the AF phase,

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

We anticipate the role of all RWCs in Tab. 1.1.

1.1.1 Antiferromagnetism in the HM

In this short section we recall the principal features of the AF phase established in the conventional HM, discussed in Sec. **??**. Before starting, let us define as in Eqns. (**??**) and (**??**) the AF and density fluctuations operators,

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{1.1}$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{1.2}$$

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalization and resymmetrization, gap complexification

Table 1.1 Relevant Wick's contractions and net effect in the AF phase. Note the similarity with Tab. ??.

which represent the two essential fermionic quadratic operators of the analytic MFT derivation. It is very useful to also note the following properties. Given the γ-wave form factors of Tab. ??, evidently both the following equations hold

$$\hat{\psi}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} = \hat{\psi}_{\mathbf{k}\sigma}$$
$$\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}} = -\varphi^{(\gamma)}_{\mathbf{k}}$$

which imply the handy property

$$\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}}\psi^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} = -\varphi^{(\gamma)}_{\mathbf{k}}\psi_{\mathbf{k}\sigma} \tag{1.3}$$

Within the pure HM, the AF phase is specified by the Ansatz (??),

$$\langle \hat{n}_{\mathbf{r}\sigma}\rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \tag{1.4}$$

which is explicitly breaking translational invariance in each spin sector, while preserving $\mathrm{U}^z(1)$ and $\mathrm{U}^c(1)$ symmetries. As reported in Tab. ??, the only RWC comes in Hartree channel, since the local two-body interaction couples operators at opposite spin index. The hamiltonian $\hat{H}$ is reduced thereby to the form of Eq. (??)

$$\hat{H}_t + \hat{H}_U \overset{\text{MFT}}{\simeq} -t\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_\sigma \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma} + \text{h.c.}\right] + nU\sum_{\mathbf{r}}\left[\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\right] - mU\sum_{\mathbf{r}}(-1)^{x+y}\left[\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] - E^{(\text{AF})}_{\text{H}/U}$$

with $E^{(\text{AF})}_{\text{H}/U}$ the contraction energy resulting from double counting terms, given explicitly in Eq. (??),

$$E^{(\text{AF})}_{\text{H}/U} = L_x L_y \times U\left(n^2 - m^2\right)$$

In reciprocal space, the hamiltonian decomposes as in Eq. (??),

$$\hat{H}_t + \hat{H}_U \overset{\text{MFT}}{\simeq} \sum_{\mathbf{k}\in\text{MBZ}}\sum_\sigma \hat{\Psi}^\dagger_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} + nU\times\hat{N} - E^{(\text{AF})}_{\text{H}/U} \quad \text{being} \quad h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta^*_{\mathbf{k}\sigma} \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma}\end{bmatrix}$$

and $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. Nambu spinorial formulation is used,

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix}\hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\end{bmatrix}$$

and the free electrons energy is simply the tight binding energy of Eq. (??) written with an explicit spin index,

$$\epsilon_{\mathbf{k}\sigma} = -2t\left(\cos k_x + \cos k_y\right)$$

which is spin-invariant. The MFT description of the model reduces to a gas of free "γ-fermions", described by the Nambu spinor of Eq. (??),

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix}\hat{\gamma}^{(+)}_{\mathbf{k}\sigma} \\ \hat{\gamma}^{(-)}_{\mathbf{k}\sigma}\end{bmatrix}$$

EV	Meaning	Equation
$\langle \hat{s}^x_{\mathbf{k}\sigma} \rangle$	$\left\langle \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma} \right\rangle$	(??)
$\langle \hat{s}^y_{\mathbf{k}\sigma} \rangle$	$i \left\langle \hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^\dagger_{\mathbf{k}\sigma} \right\rangle$	(??)
$\langle \hat{s}^z_{\mathbf{k}\sigma} \rangle$	$\langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle$	(??)

Table 1.2 List of the various expectation values (EVs) of each pseudospin component, their meaning in terms of fermionic operators and the reference equations from the pure HM.

where

$$W_{\mathbf{k}\sigma} = \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix} \quad \text{and} \quad \sin 2\theta_{\mathbf{k}\sigma} \equiv \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}$$

These fermions populate the two bands $\pm E_{\mathbf{k}\sigma} = \sqrt{\epsilon^2_{\mathbf{k}\sigma} + |\Delta_{\mathbf{k}\sigma}|^2}$. The entire system is mapped onto an ensemble of pseudo-spins, each subject to a pseudo-field, as in Fig. 1.1. To diagonalize the system essentially means to anti-align each pseudo-spin with the z axis. Each pseudospin component expectation value (EV) is described by a simple equation and has a precise meaning, listed in Tab. 1.2.

1.1.2 Extension to the EHM

Consider now the non-local interaction $\hat{H}_V$. Being the RWCs the same as for the normal phase, the discussion is essentially the same of Chap. **??**, with the only relevant difference being the Ansatz we use in Eq. (**??**), substituted by the commensurate oscillatory Ansatz of Eq. (1.4). By itself, even if it may seem the contrary, the Ansatz does not imply C_4 symmetry preservation. It is possible to reduce point group symmetry having still a rotation-symmetric magnetization, as well as it was possible to do in the normal phase with keeping a uniform density. As for the normal phase, by design the MFT solution to the model for the given phase is described by a redefinition of the various physical quantities and mathematical objects,

$$\epsilon_{\mathbf{k}\sigma} \to \tilde{\epsilon}_{\mathbf{k}\sigma} \qquad E_{\mathbf{k}\sigma} \to \tilde{E}_{\mathbf{k}\sigma} \qquad \Delta_{\mathbf{k}\sigma} \to \tilde{\Delta}_{\mathbf{k}\sigma} \qquad \mu \to \tilde{\mu}$$

The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}\sigma} \equiv \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + |\tilde{\Delta}_{\mathbf{k}\sigma}|^2}$$

and the following relations hold (the *tilde* sign indicates the renormalized quantity):

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^x \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}\sigma} \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{1.5}$$
$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^y \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}\sigma} \sin 2\tilde{\zeta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{1.6}$$
$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Psi}_{\mathbf{k}\sigma} \rangle = \cos 2\tilde{\theta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{1.7}$$

equivalent to the ones for the pure HM, Eqns. (**??**), (**??**) and (**??**). Moving to the the tilted frame, the z component of the pseudospin is just a thermal superposition of the "up" and "down" states

$$\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) - f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \tag{1.8}$$

Let us define the (renormalized) thermal factors as in Eq. (**??**),

$$\tilde{\mathrm{Th}}^{(\pm)}_{\mathbf{k}\sigma}[\beta, \tilde{\mu}] \equiv f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \pm f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \tag{1.9}$$

The angles satisfy:

$$\sin 2\tilde{\xi}_{\mathbf{k}\sigma} = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \cos 2\tilde{\xi}_{\mathbf{k}\sigma} = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \sin 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{|\tilde{\Delta}_{\mathbf{k}\sigma}|}{\tilde{E}_{\mathbf{k}\sigma}} \quad \cos 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tag{1.10}$$

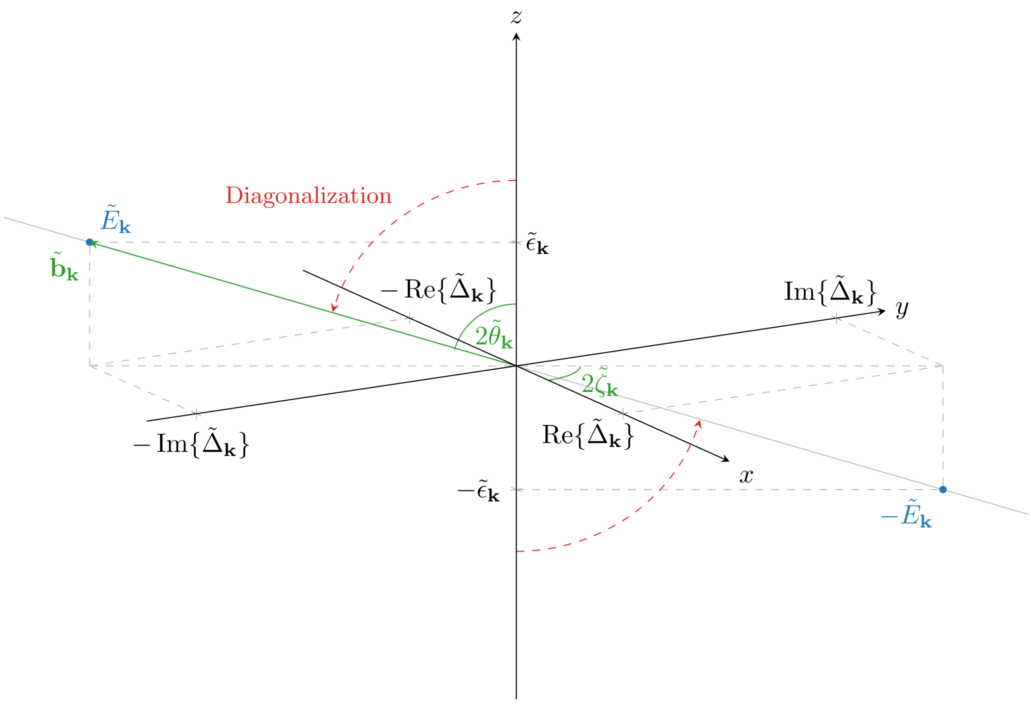

Figure 1.1 Sketch of the diagonalization of the pseudo-spin problem. The dashed line represents the diagonalization given by the z-axis alignment with the field direction. In the top-right side of the picture are listed the expectation values of the original Nambu spinor. For simplicity everywhere the index $\sigma =\uparrow$ has been omitted.

The physical behaviour is the same as for the pure HM. In terms of MFT discussion, the AF phase is similar enough to the normal phase of Chap. **??**: the RWCs are the same of Sec. **??**, listed in Tabs. **??** and **??**. All three Hartree contractions, responsible in the normal phase for the chemical potential shift, and the s.s. Fock contraction, that originated the hopping renormalization. Let us start by the Hartree terms: due to the "staggered" nature of the commensurate AF phase, their role is here more complex. Then we will move to the Fock term, discussing the MFT approximations effect on bands. Next sections are basically an extension of the calculations of Chap. **??**.

1.2 Non-local Hartree effects: chemical potential renormalization

As in Eq. (**??**), the same-spin and opposite-spin non-local Hartree terms are

$$\hat{H}_V \simeq \overbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\hat{n}_{j\sigma}+\hat{n}_{i\sigma}\langle\hat{n}_{j\sigma}\rangle-\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle\right]}^{\text{s.s.}} \\ \underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{n}_{i\sigma}\rangle\hat{n}_{j\bar{\sigma}}+\hat{n}_{i\sigma}\langle\hat{n}_{j\bar{\sigma}}\rangle-\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\bar{\sigma}}\rangle\right]}_{\text{o.s.}}+(\text{all the rest})$$

Let $i \to \mathbf{r} = (x, y)$ and $j \to \mathbf{r}' = (x', y')$. We use Eq. (1.4), which describes an oscillatory density for each spin sector. This Ansatz is composed of two parts: the offset n, and the oscillatory part m. The algebraic derivation relative to the n part is, of course, the same as for the normal phase. We now treat separately the "operatorial" part of the braced terms of Eq. (**??**),

$$-V\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\langle\hat{n}_{\mathbf{r}\sigma}\rangle\hat{n}_{\mathbf{r}'\sigma}+\hat{n}_{\mathbf{r}\sigma}\langle\hat{n}_{\mathbf{r}'\sigma}\rangle\right]-V\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\langle\hat{n}_{\mathbf{r}\sigma}\rangle\hat{n}_{\mathbf{r}'\bar{\sigma}}+\hat{n}_{\mathbf{r}\sigma}\langle\hat{n}_{\mathbf{r}'\bar{\sigma}}\rangle\right] \tag{1.11}$$

as well as the its scalar counterpart

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\sigma} \rangle + V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\overline{\sigma}} \rangle \tag{1.12}$$

which accounts for the energy shift to be accounted when performing Wick's contractions.

1.2.1 Hartree channel in the AF phase: analytic derivation

Let us start from the operator of Expr. (1.11). By inserting the Ansatz of Eq. (??), we get

$$\overbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma}] + mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{\delta_{\sigma=\uparrow}} \left[(-1)^{x+y} \hat{n}_{\mathbf{r}'\sigma} + (-1)^{x'+y'} \hat{n}_{\mathbf{r}\sigma} \right]}^{\text{s.s.}}$$
$$\underbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}'\overline{\sigma}} + \hat{n}_{\mathbf{r}\sigma}] + mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[(-1)^{x+y+\delta_{\sigma=\uparrow}} \hat{n}_{\mathbf{r}'\overline{\sigma}} + (-1)^{x'+y'+\delta_{\overline{\sigma}=\uparrow}} \hat{n}_{\mathbf{r}\sigma} \right]}_{\text{o.s.}}$$

For a square lattice, if $\mathbf{r} = (x, y)$ and $\mathbf{r}' = (x', y')$ are NNs, evidently

$$(-1)^{x'+y'} = (-1)^{x+y+1}$$
$$(-1)^{\delta_{\overline{\sigma}=\uparrow}} = (-1)^{\delta_{\sigma=\uparrow}+1}$$

We obtain

$$\overbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\sigma}]}^{\text{s.s. (density)}} \overbrace{- mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{x+y+\delta_{\sigma=\uparrow}} [\hat{n}_{\mathbf{r}\sigma} - \hat{n}_{\mathbf{r}'\sigma}]}^{\text{s.s. (magnetization)}}$$
$$\underbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\overline{\sigma}}]}_{\text{o.s. (density)}} + \underbrace{mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{x+y+\delta_{\sigma=\uparrow}} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\overline{\sigma}}]}_{\text{o.s. (magnetization)}} \tag{1.13}$$

In the above expressions the various contribution have been separated in "density" contributions and "magnetization" contributions. Let us deal with these separately.

Density terms. Consider the s.s. and o.s. (density) terms of Expr. (1.13),

$$-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\sigma}] - nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\overline{\sigma}}] \tag{1.14}$$

The discussion is identical to the one of Sec. ?? for the normal phase, with the renormalization of μ being the same of Eq. (??).

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{1.15}$$

Within this equation we are also already implicitly considering the μ shift effect due to $\hat{H}_U$., discussed both in Sec. ?? and Sec. ??.

Magnetization terms. The s.s. and o.s. (magnetization) terms of Expr. (1.13) perfectly cancel out:

$$mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{x+y+\delta_{\sigma=\uparrow}} [(\hat{n}_{\mathbf{r}\sigma} - \hat{n}_{\mathbf{r}'\sigma}) - (\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\overline{\sigma}})] = 0$$

thus the only net effect of the Hartree RWCs affects μ. This is somewhat to be expected: in the context of the HM, the gap function emerges in its purely s-wave structure from this precise computation on Hartree terms. However, the $\hat{H}_V$ non-local interaction cannot provide a pure s-wave potential component. Effects on the gap function are to be observed somewhere else.

Let us now move to the last part of the Hartree effects on the AF phase originated by $\hat{H}_V$, which is, the energy shift due to double counting terms originated by MFT decomposition of the quartic interactions $\hat{H}_U$ and $\hat{H}_V$.

1.2.2 Hartree contraction energy

Consider the scalar Expr. (1.12): it's the sum of two parts, the first coming from s.s. sector, the second from o.s. sector. In computing the contractyion energy from Hartree channel, being the operatorial result identical to the one obtained for the normal phase, we expect the saeme to happen for double counting terms.

s.s. sector double counting energy. For the s.s. part, explicitating the two spin sectors, we get

$$V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle \hat{n}_{\mathbf{r}\sigma}\rangle\langle \hat{n}_{\mathbf{r}'\sigma}\rangle = \overbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\left[(n+(-1)^{x+y}m)(n+(-1)^{x'+y'}m)\right]}^{\sigma=\uparrow} + \underbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\left[(n-(-1)^{x+y}m)(n-(-1)^{x'+y'}m)\right]}_{\sigma=\downarrow}$$

Evidently the mixed terms (those of order $\mathcal{O}(n)\times\mathcal{O}(m)$) cancel out, and considering that the sign factor is necessarily alternating on neighboiring site, the remainder is just

$$V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\sigma}\rangle = V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}2(n^2-m^2)$$

o.s. sector double counting energy. Proceeding identically on the o.s. sector, we get

$$V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\langle \hat{n}_{\mathbf{r}\sigma}\rangle\langle \hat{n}_{\mathbf{r}'\overline{\sigma}}\rangle = \overbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\left[(n+(-1)^{x+y}m)(n-(-1)^{x'+y'}m)\right]}^{\sigma=\uparrow} + \underbrace{V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\left[(n-(-1)^{x+y}m)(n+(-1)^{x'+y'}m)\right]}_{\sigma=\downarrow}$$

which gives immediately

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\overline{\sigma}}\rangle = V\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}2(n^2+m^2)$$

Hence, the total "contraction energy", given by the sum of both contributions, is identical to the normal one of Eq. (**??**) indeed:

$$E^{(\mathrm{AF})}_{\mathrm{H}/V} = -2V\sum_{\langle ij\rangle}\sum_{\sigma}n^2 = 4n^2V\times\frac{z}{2}L_xL_y \tag{1.16}$$

the minus sign coming from the fact that we are subtracting this energy shift.

1.3 Fock effects: gap and bands renormalization

We explored thoroughly the Hartree contractions effects on the MFT hamiltonian. Let us now move to the effects originated by Fock contractions. We can use the very beginning of Sec. **??** derivation, up to Eq. (**??**),

$$\begin{aligned} &V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right] \\ &\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}-E^{(\mathrm{N})}_{\mathrm{F}/V}\end{aligned} \tag{1.17}$$

where $E^{(\mathrm{AF})}_{\mathrm{F}/V}$ is defined as in Eq. (**??**), with the expectation value being computed on the AF (thermal) ground-state. As before, we discuss the operatorial part and the scalar part separately.

1.3.1 Fock channel in the AF phase: analytic derivation

The commensurate AF phase is essentially described by two EVs being in principle non-zero,

$$\left|\langle \hat{n}_{\mathbf{k}\sigma}\rangle\right| \geq 0 \qquad \left|\langle \hat{\psi}_{\mathbf{k}\sigma}\rangle\right| \geq 0$$

Because of this, for the conventional HM the AF ground state is a degenerate Fermi gas of $\gamma^{(\pm)}$ quasiparticles, and for null temperature acquires the simple form of Eq. (**??**). Assuming the new ground state to be similar and going back to Eq. (1.17), this implies only two contributions are non-zero:

$$\mathbf{k} = -\mathbf{k}' \quad \text{or} \quad \mathbf{k}+\boldsymbol{\pi} = -\mathbf{k}'$$

The first case gives us the normal fermionic EV, $\langle \hat{n}_{\mathbf{k}\sigma}\rangle$, the second the staggered AF order parameter, $\langle \hat{\psi}_{\mathbf{k}\sigma}\rangle$. Then Eq. (1.17) is reduced to:

$$\begin{aligned}(\cdots) \overset{\mathcal{F}}{=} \frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}\left[\cos\left(2k_x\right)+\cos\left(2k_y\right)\right]&\\ \Big[\underbrace{\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}\sigma}}_{\text{Diagonal terms}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}}_{\text{Off-diagonal terms}}\Big]&\end{aligned} \tag{1.18}$$

Now, the above equation presents *diagonal* and *off-diagonal* terms.

Diagonal terms. The diagonal terms of Eq. (1.18) are simple density interactions with the mean density field. Consider Fig. **??**: density at vector $\mathbf{q} \equiv \mathbf{K}+\mathbf{k}$ interacts with the mean density at vector $\mathbf{q}' \equiv \mathbf{K}-\mathbf{k}$. These variables are depicted in Fig. **??**. Apply in the diagonal part of Eq. (1.18) this variables change (which is the same of Eq. (**??**))

$$\begin{aligned}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}&\left[\cos\left(2k_x\right)+\cos\left(2k_y\right)\right]\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}\sigma}\\ &= \frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_{\sigma}\left[\cos\left(\delta q_x\right)+\cos\left(\delta q_y\right)\right]\langle \hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle \hat{c}^\dagger_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma}\\ &= \sum_{\gamma,\sigma}\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}}\varphi^{(\gamma)}_{\mathbf{q}}\langle \hat{n}_{\mathbf{q}\sigma}\rangle\right]\times V\sum_{\mathbf{q}'}\varphi^{(\gamma)*}_{\mathbf{q}'}\hat{n}_{\mathbf{q}'\sigma}\end{aligned} \tag{1.19}$$

where in the last passage we substituted the decomposition of Eq. (**??**). Define as in Eq. (**??**) the EV

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{n}_{\mathbf{q}\sigma}\rangle \tag{1.20}$$

The expression in square brackets of Eq. (1.19) is just its γ component, and is expanded in

$$\begin{aligned}u^{(\gamma)}_{\sigma} &= \frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle \hat{n}_{\mathbf{q}\sigma}\rangle\\ &= \frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[\varphi^{(\gamma)}_{\mathbf{q}}\langle \hat{n}_{\mathbf{q}\sigma}\rangle + \varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}}\langle \hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle\right]\\ &= \frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\varphi^{(\gamma)}_{\mathbf{q}}\left[\langle \hat{n}_{\mathbf{q}\sigma}\rangle - \langle \hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle\right]\\ &= \frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle \hat{\Psi}^\dagger_{\mathbf{q}\sigma}\tau^z\hat{\Psi}_{\mathbf{q}\sigma}\rangle\\ &= -\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{MBZ}}\varphi^{(\gamma)}_{\mathbf{q}}\frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta,\tilde{\mu}]\end{aligned} \tag{1.21}$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_\mathbf{k}}{\tilde{E}_\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_\mathbf{k}[\beta, \tilde{\mu}]$	Fig. **??**
$d_{x^2-y^2}$-wave	$u^{(d)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x - \cos k_y) \frac{\tilde{\epsilon}_\mathbf{k}}{\tilde{E}_\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_\mathbf{k}[\beta, \tilde{\mu}]$	Fig. **??**

Table 1.3 | Symmetry and spin resolved self-consistency equations for the harmonics of $u_\mathbf{k}$.

where, in the last passage, Eqns. (1.7), (1.8) and (1.9) have been used. Now, the AF phase we realize does not break inversion symmetry (and as a consequence, neither C_2): this implies that, whatever the shape of the renormalized bands, it must be

$$\tilde{\epsilon}_{\mathbf{q}\sigma} = \tilde{\epsilon}_{-\mathbf{q}\sigma} \quad \Longrightarrow \quad u^{(p_\ell)}_\sigma = 0 \quad \text{with} \quad \ell \in \{x, y\}$$

The renormalized bands need to be symmetric functions of the moment. Hence for $\gamma \in \{p_x, p_y\}$ the above sum vanishes. The remaining s^* and d symmetries are legitimate HFPs with relative self-consistency equations, reported in Tab. 1.3 using spin degeneracy.

Recognizing the harmonics, Eq. (1.19) becomes

$$\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_\sigma [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}$$
$$= V \sum_{\mathbf{q},\sigma} \left[u^{(s^*)}_\sigma (\cos q_x + \cos q_y) + u^{(d)}_\sigma (\cos q_x - \cos q_y) \right] \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma}$$

These two factors, together, define the full bands renormalization

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u^{(s^*)}_\sigma V (\cos k_x + \cos k_y) + u^{(d)}_\sigma V (\cos k_x - \cos k_y) \tag{1.22}$$

as was for the normal phase in Eq. (**??**). In Sec. **??** we discussed the appearance of anisotropic hopping with d-wave extension of the bands: here the theory predicts something identical. These RMPs are, as before, controlled linearly by V and in principle can lead to relevant changes in the kinetic sector of the hamiltonian.

Off-diagonal terms. Consider the off-diagonal terms of Eq. (1.18). These contribute instead to the full gap, being out of diagonal in the 2×2 hamiltonian matrix. In terms of the pseudospin picture, they account for a transverse field. Define $\mathbf{q}$, $\mathbf{q}'$ as in Fig. **??**,

$$\mathbf{q} \equiv \mathbf{K} + \mathbf{k} \qquad \mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$$

and rewrite as for Eq. (1.19)

$$-\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_\sigma [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= -\sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_\mathbf{q} \langle \hat{\psi}^\dagger_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{\psi}^\dagger_{\mathbf{q}'\sigma}$$

Now shift the first sum by an AF vector $\boldsymbol{\pi}$ defining $\mathbf{k} + \boldsymbol{\pi} \equiv \mathbf{q}$, and use Eq. (1.3) as well as reciprocal space periodicity by a BZ,

$$\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_\mathbf{q} \langle \hat{\psi}^\dagger_{\mathbf{q}\sigma} \rangle = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}} \langle \hat{\psi}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle$$
$$= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)}_\mathbf{k} \langle \hat{\psi}_{\mathbf{k}\sigma} \rangle$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$v^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \frac{\mathrm{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_\mathbf{k}[\beta, \tilde{\mu}]$	Fig. ??
p_x-wave	$v^{(p_x)} = \frac{i\sqrt{2}}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \sin k_x \frac{\mathrm{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_\mathbf{k}[\beta, \tilde{\mu}]$	Fig. ??
p_y-wave	$v^{(p_y)} = \frac{i\sqrt{2}}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \sin k_y \frac{\mathrm{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_\mathbf{k}[\beta, \tilde{\mu}]$	Fig. ??
$d_{x^2-y^2}$-wave	$v^{(d)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x - \cos k_y) \frac{\mathrm{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_\mathbf{k}[\beta, \tilde{\mu}]$	Fig. ??

Table 1.4 | Symmetry and spin resolved self-consistency equations for the harmonics of $v_\mathbf{k}$.

which finally gives

$$-\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_\mathbf{q} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{\psi}^\dagger_{\mathbf{q}'\sigma} \tag{1.23}$$

Proceed as done previously: define the expectation value

$$v_{\mathbf{q}\sigma} \equiv i \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \tag{1.24}$$

(which is just the AF order parameter with a complex factor) and its γ-wave component, with $\gamma \in \{s^*, d, p_x, p_y\}$:

$$\begin{aligned} v^{(\gamma)}_\sigma &= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_\mathbf{q} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \\ &= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \left[\varphi^{(\gamma)}_\mathbf{q} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle + \varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} \langle \hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\ &= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_\mathbf{q} \left[\langle \hat{\psi}_{\mathbf{q}\sigma} \rangle - \langle \hat{\psi}^\dagger_{\mathbf{q}\sigma} \rangle \right] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_\mathbf{q} \langle \hat{\Psi}^\dagger_{\mathbf{q}\sigma} \tau^y \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi^{(\gamma)}_\mathbf{q} \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}] \end{aligned} \tag{1.25}$$

where we made use of Eq. (1.3), the relations of Tab. 1.2 as well as Eq. (1.6). There is no particular reason to exclude one component or another. In Tab. 1.4 the four derived self-consistency equations are reported (already ignoring spin index).

From the definitions of the structure factors of Tab. ??, it's immediate to see that:

$$v^{(s^*)}_\sigma, v^{(d)}_\sigma \in \mathbb{R} \qquad v^{(p_x)}_\sigma, v^{(p_y)}_\sigma \in i\mathbb{R}$$

Getting back to Eq. (1.23) and inserting these components, we get

$$-\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= -iV \sum_{\gamma,\sigma} v^{(\gamma)}_\sigma \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)*}_\mathbf{q} \hat{\psi}^\dagger_{\mathbf{q}\sigma} \tag{1.26}$$

We can expand on the MBZ as

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= -iV\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}^\dagger_{\mathbf{q}\sigma}+v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}+\boldsymbol{\pi}}\hat{\psi}^\dagger_{\mathbf{q}+\boldsymbol{\pi}\sigma}\right]$$

and use Eq. (1.3), exchanging the terms positions

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= iV\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}_{\mathbf{q}\sigma}-v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}^\dagger_{\mathbf{q}\sigma}\right] \tag{1.27}$$

Now: recalling Tab. **??**, for the four symmetries we are considering, the form factors are real for symmetries s^* and d, and purely imaginary for p_x and p_y. As we mentioned before, the same holds for the $v^{(\gamma)}_\sigma$ harmonics. Then

$$iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}} \in i\mathbb{R} \tag{1.28}$$

and to conjugate a purely imaginary number it suffices to put a minus sign in front of it,

$$-iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}} = \left(iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\right)^*$$

Therefore, we conclude

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos(2k_x)+\cos(2k_y)]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= V\sum_{\gamma,\sigma}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left[iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}_{\mathbf{q}\sigma}+\mathrm{h.c.}\right] \tag{1.29}$$

which is obvious, being the hamiltonian operator hermitian from the beginning.

1.3.2 Fock contraction energy

Let us now get back to $E^{(\mathrm{AF})}_{\mathrm{F}/V}$, the double counting energy of Eq. (1.17). As said before, its definition is the same as for the normal phase, Eq. (**??**), with the averaging operation being performed on the AF ground state:

$$E^{(\mathrm{AF})}_{\mathrm{F}/V} \equiv \frac{V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}[\cos(\delta k_x)+\cos(\delta k_y)]\,\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\langle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\rangle \tag{1.30}$$

Proceeding exactly as in Eq. (1.18) and making the change of variables of Fig. **??**, we get

$$E^{(\mathrm{AF})}_{\mathrm{F}/V} = \frac{V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_{\sigma}[\cos(\delta q_x)+\cos(\delta q_y)]\left[\underbrace{\langle\hat{n}_{\mathbf{q}\sigma}\rangle\langle\hat{n}_{\mathbf{q}'\sigma}\rangle}_{\text{Diagonal terms}}-\underbrace{\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle\langle\hat{\psi}_{\mathbf{q}'\sigma}\rangle}_{\text{Off-diagonal terms}}\right]$$

Apply the decomposition of Eq. (**??**) and use Eqns. (1.20), (1.24),

$$E^{(\mathrm{AF})}_{\mathrm{F}/V} = \frac{V}{2L_xL_y}\sum_{\gamma,\sigma}\sum_{\mathbf{q},\mathbf{q}'}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}\left[u_{\mathbf{q}\sigma}u_{\mathbf{q}'\sigma}+v_{\mathbf{q}\sigma}v_{\mathbf{q}'\sigma}\right] \tag{1.31}$$

Now: it holds

$$u_{\mathbf{q}\sigma} = \langle\hat{n}_{\mathbf{q}\sigma}\rangle \quad\Longrightarrow\quad u_{\mathbf{q}\sigma} = u^*_{\mathbf{q}\sigma}$$

and then

$$\begin{aligned}\frac{1}{(L_xL_y)^2}\sum_{\mathbf{q},\mathbf{q}'}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}u_{\mathbf{q}\sigma}u_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}u_{\mathbf{q}\sigma}\right]\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}u^*_{\mathbf{q}'\sigma}\right]\\ &= \left|u^{(\gamma)}_\sigma\right|^2\end{aligned}$$

At the same time, being $\hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma} = \hat{\psi}^\dagger_{\mathbf{q}\sigma}$,

$$v_{\mathbf{q}\sigma} = i\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle \quad\Longrightarrow\quad v_{\mathbf{q}+\boldsymbol{\pi}\sigma} = -v^*_{\mathbf{q}\sigma}$$

Since the form factors satisfy $\varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} = -\varphi^{(\gamma)}_{\mathbf{q}}$, we get:

$$\begin{aligned}\frac{1}{(L_xL_y)^2}\sum_{\mathbf{q},\mathbf{q}'}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}v_{\mathbf{q}\sigma}v_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}v_{\mathbf{q}\sigma}\right]\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}v_{\mathbf{q}'\sigma}\right]\\ &= v^{(\gamma)}_\sigma\left[\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{k}+\boldsymbol{\pi}}v_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]\\ &= v^{(\gamma)}_\sigma\left[\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{k}}v^*_{\mathbf{k}\sigma}\right]\\ &= \left|v^{(\gamma)}_\sigma\right|^2\end{aligned}$$

where in the second passage we employed the change of variables $\mathbf{q}' = \mathbf{k}+\boldsymbol{\pi}$ and used BZ periodicity to map the original sum onto a new sum for $\mathbf{k} \in$ BZ. Inserting the above results in Eq. (1.31), this last becomes:

$$E^{(\mathrm{AF})}_{\mathrm{F}/V} = L_xL_y \times \frac{V}{2}\sum_{\gamma,\sigma}\left[\left|u^{(\gamma)}_\sigma\right|^2 + \left|v^{(\gamma)}_\sigma\right|^2\right] \tag{1.32}$$

Interestingly enough, the new HFPs amplitudes contribute positively to the contraction energy. This fact is non-trivial and has consequences we will discuss in Sec. [Missing].

1.4 Full MFT hamiltonian and gap complexification

Finally, it is time to collect everything in a single hamiltonian operator. The MFT model is approximated by:

$$\begin{aligned}\hat{H} - \mu\hat{N} \simeq \sum_\sigma\sum_{\mathbf{k}\in\mathrm{MBZ}}\tilde{\epsilon}_{\mathbf{k}\sigma}\left[\hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right] - \sum_\sigma\sum_{\mathbf{k}\in\mathrm{MBZ}}\left[\tilde{\Delta}_{\mathbf{k}\sigma}\hat{\psi}_{\mathbf{k}\sigma} + \tilde{\Delta}^*_{\mathbf{k}\sigma}\hat{\psi}^\dagger_{\mathbf{k}\sigma}\right]\\ - \tilde{\mu}\hat{N} - \left[E^{(\mathrm{AF})}_{\mathrm{H}/U} + E^{(\mathrm{AF})}_{\mathrm{H}/V} + E^{(\mathrm{AF})}_{\mathrm{F}/V}\right]\end{aligned}$$

where the renormalized bands are given by:

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} + V\sum_{\gamma\in\{s^*,d\}}u^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{k}}$$

and the full gap function by

$$\tilde{\Delta}_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma} - iV\sum_\gamma v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{k}} \tag{1.33}$$

The gap correction is purely imaginary, compared to the HM gap $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}}mU$. This has two obvious consequences: being no other contribution to the x pseudospin projections other than the pure HM ones, spin antisymmetry is preserved within such direction

$$\langle\hat{s}^x_{\mathbf{k}\sigma}\rangle = -\langle\hat{s}^x_{\mathbf{k}\bar{\sigma}}\rangle$$

which is relevant in the computation of the physical magnetization, which (as is proven in Eq. (**??**)) is defined

$$\begin{aligned} m &= \frac{1}{2L_xL_y}\sum_{\mathbf{r}}(-1)^{x+y}\langle \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\rangle \\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle \hat{s}^x_{\mathbf{k}\uparrow} - \hat{s}^x_{\mathbf{k}\downarrow}\right\rangle \\ &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle \hat{s}^x_{\mathbf{k}\uparrow}\right\rangle \end{aligned}$$

A proof of the above derivation is given in Sec. **??**. Moreover, the net effect $\hat{H}_V$ is a "gap complexification". In other words,

$$\left|\,\mathrm{Im}\{\Delta_{\mathbf{k}\sigma}\}\right| \geq 0 \quad\implies\quad \left|\,\langle s^y_{\mathbf{k}\sigma}\rangle\,\right| \geq 0$$

while for the HM the y pseudospin component was identically null.

1.4.1 The relative phase in the zero temperature half-filled ground state

In Sec. **??**, we derive analytically the AF ground state on the conventional HM at zero temperature, as a simple Bloch's transform of a polarized (tilted) pseudospins collective pure state. Eq. (**??**) reports the final form of said ground-state. Said quantum state is in general pretty similar to the conventional BCS state.

At half-filling and zero temperature, it is rather easy to extend that state to the new complexified gap. By performing an identical calculation as in Sec. **??** we here omit, it's easy to show that the new ground state for the EHM in the AF phase is identical, with RMPs in the various amplitudes and a relative phase taking care of the gap function complex phase,

$$|\Psi\rangle = \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\left[\sin\tilde{\theta}_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}}\cos\tilde{\theta}_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right]|\Omega_{\mathbf{k}\sigma}\rangle \tag{1.34}$$

which is identical to Eq. (**??**), apart from renormalizations. Let $|\mathbf{k}\sigma\rangle \equiv \hat{c}^\dagger_{\mathbf{k}\sigma}\,|\Omega_{\mathbf{k}\sigma}\rangle$. Then we can write

$$|\Psi\rangle = \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}}\left[\sin\tilde{\theta}_{\mathbf{k}\sigma} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}}\cos\tilde{\theta}_{\mathbf{k}\sigma}\hat{\psi}_{\mathbf{k}\sigma}\right]|\mathbf{k}\sigma\rangle$$

having we used $|\Omega_{\mathbf{k}\sigma}\rangle = \{\hat{c}_{\mathbf{k}\sigma}, \hat{c}^\dagger_{\mathbf{k}\sigma}\}\,|\Omega_{\mathbf{k}\sigma}\rangle$.

The angles appearing in the above equations are the same of Eq. (1.10), which acquire a physical meaning: the relative phase $2\tilde{\zeta}_{\mathbf{k}\sigma}$ is the gap function angle in complex plane, since

$$\begin{aligned} \langle\Psi|\hat{s}^x_{\mathbf{k}\sigma}|\Psi\rangle &= \langle\Psi|\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma}|\Psi\rangle \\ &= \sin\tilde{\theta}_{\mathbf{k}\sigma}\cos\tilde{\theta}_{\mathbf{k}\sigma}\left(e^{-2i\tilde{\zeta}_{\mathbf{k}\sigma}} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}}\right) \\ &= \sin 2\tilde{\theta}_{\mathbf{k}\sigma}\cos 2\tilde{\zeta}_{\mathbf{k}\sigma} \end{aligned}$$

$$\begin{aligned} \langle\Psi|\hat{s}^y_{\mathbf{k}\sigma}|\Psi\rangle &= i\,\langle\Psi|\hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^\dagger_{\mathbf{k}\sigma}|\Psi\rangle \\ &= i\sin\tilde{\theta}_{\mathbf{k}\sigma}\cos\tilde{\theta}_{\mathbf{k}\sigma}\left(e^{-2i\tilde{\zeta}_{\mathbf{k}\sigma}} - e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}}\right) \\ &= \sin 2\tilde{\theta}_{\mathbf{k}\sigma}\sin 2\tilde{\zeta}_{\mathbf{k}\sigma} \end{aligned}$$

These equations are coherent with Eqns. (1.5), (1.6), thus the involved angles apart from irrelevant 2π shifts coincide.

1.4.2 AF phase flavours and spin degeneracy

The AF phase we are studying is not allowed to break inversion symmetry, a core property of the system which is respected from all our approximations. This translates in the simple requirement that:

$$\tilde{E}_{\mathbf{k}\sigma} \stackrel{!}{=} \tilde{E}_{-\mathbf{k}\sigma}$$

Phase	HFPs
sAF	$\{m, u^{(s^*)}, u^{(d)}, v^{(s^*)}, v^{(d)}\}$
aAF	$\{m, u^{(s^*)}, u^{(d)}, -iv^{(p_x)}, -iv^{(p_y)}\}$

Table 1.5 Set of HFP for the two possible realization of the AF phase, respectively a symmetric and an antisymmetric imaginary part of the gap.

We know that:

$$\tilde{E}_{\mathbf{k}\sigma} = \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + \mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\} + \mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}$$

The renormalized bare bands as well as the real part of the gap satisfy the inversion symmetry requirement,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \tilde{\epsilon}_{-\mathbf{k}\sigma} \qquad \mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \mathrm{Re}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

hence inversion symmetry is respected only if

$$\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \pm\,\mathrm{Im}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

which is: either the imaginary part of the gap is inversion symmetric, or is antisymmetric. Based on this trivial consideration, we are able two discard two of the four self-consistency equations for the v-related HFPs listed in Tab. 1.4: s^* and d-wave symmetries HFPs are nonzero if $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-even; p_x and p_y-wave symmetries HFPs are nonzero if $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-odd. The two options cannot coexist, and separate simulations can be implemented in the two symmetry sectors. We name the two possible symmetries “sAF” (symmetric antiferromagnet) and “aAF” (antisymmetric antiferromagnet). The set of HFPs then restricts in these AF structures, and is reported for both cases in Tab. 1.5.

Moreover, from now on we will omit the σ index, working just in the $\uparrow$ sector. As said in Sec. 1.1, the AF phase does not break $\mathbb{Z}_2$ spin inversion symmetry, being the SDW ordering generated by a staggered but globally symmetric spin density distribution among sites. The MFT hamiltonian is invariant by a $\sigma \leftrightarrow \overline{\sigma}$ exchange, thus all measurable physical quantities must be. What is unmeasurable and has no effect on Physics is free to be σ-dependent, as for the sign of $\mathrm{Re}\{\Delta_{\mathbf{k}\sigma}\}$. Since all it’s relevant of both the complex components of the gap function is their magnitude, we can safely treat σ as a pure degeneracy. This gives a simple equation for the last HFP,

$$\begin{aligned} m &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \langle \hat{s}^x_{\mathbf{k}\uparrow} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}] \end{aligned} \tag{1.35}$$

which is the actual physical magnetization. From the first line we see that it can be interpreted as the average transverse pseudospin value.

1.5 Stability of the AF phase

[Re-write and move up.]

The discussion above is formally correct and self-consistent, thus a HF algorithm can be sketched to extract the HFPs. However we cannot proceed without considering the instability effects of Sec. **??**. As is known (and we would have liked to notice earlier) the AF-SSB at wavevector $\boldsymbol{\pi}$ we are studying here, responsible of halving the BZ down to the MBZ, can produce a stable phase on pure Hubbard lattices only at half filling [15]. For the EHM the situation is even worse than that: repeating the simple Linear Reponse Theory (LRT) computations of Sec. **??** and mimicking Eq. (**??**), we get a proper response function

$$\tilde{\chi}_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega) = \frac{\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)}{1 - [U - 2V(\cos k_x + \cos k_y)]\,\chi_{\hat{\mathbf{S}}^-\hat{\mathbf{S}}^+}(\mathbf{k},\omega)} \tag{1.36}$$

[Double check the red part and conclude this paragraph...].

As is well documented in literature [17], AF can establish in a wide range of spatial structures with particular shapes, for example stripes. In order to extract the precise AF phase, we should

1. Choose an initializer wavevector, say $\mathbf{k} = \boldsymbol{\pi}$;
2. Solve self-consistently for the HFPs at $\mathbf{k}$;
3. Compute the free energy F as a functional of the wavevector around $\mathbf{k}$;
4. Look for gradient descent of the free energy, $\boldsymbol{\nabla} F < 0$;
5. If the gradient descent is found, update $\mathbf{k} \to \mathbf{k} + \delta\mathbf{k}$ and repeat from step 2, otherwise halt.

Of course the sketched procedure can lead to a lot of complications, first the fact that at incommensurate wavevector $\mathbf{k} \neq \boldsymbol{\pi}$ the unit cell is not as simple as a pair of sites, but can become much larger requiring more refined computational solutions.

In this text we will show the self consistent result for AF-SSB at wavevector $\boldsymbol{\pi}$ even at finite doping as a first step in the above sketched procedure. The presented results are self-consistent, but unfortunately deeply unstable and unphysical at finite doping, a situation where an incommensurate and insulating AF phase is preferred to this commensurate metallic counterpart. Nevertheless we present them anyway.

Let us gather some more informations on the HFPs we identified.[To be continued. . .]

1.6 AF free energy density

In order to derive the expression of the free energy density for the AF phase, we proceed precisely as in Sec. ??. The only notable difference is in entropy: Eq. (??) is written for two opposite bands on MBZ

$$s = s^{(+)} + s^{(-)}$$

with

$$s^{(\pm)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) \ln \Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \ln f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big] \tag{1.37}$$

Consider also that we are computing free energy on the gran-canonical hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[(\tilde{E}_{\mathbf{k}} - \tilde{\mu})\hat{\gamma}^{(+)\dagger}_{\mathbf{k}\sigma}\hat{\gamma}^{(+)}_{\mathbf{k}\sigma} - (\tilde{E}_{\mathbf{k}} + \tilde{\mu})\hat{\gamma}^{(-)\dagger}_{\mathbf{k}\sigma}\hat{\gamma}^{(-)}_{\mathbf{k}\sigma}\Big] + E^{(\mathrm{AF})}_{\mathrm{c}}$$

being $E^{(\mathrm{AF})}_{\mathrm{c}}$ the total AF contraction energy. Note that on the left-hand side we have μ and on the right-hand side its RMP counterpart $\tilde{\mu}$.

With identical meaning of the notation with respect to Sec. ??, the free energy contributions are:

1. Quasiparticles energy, taking care of spin degeneracy:

$$e^{(\mathrm{AF})}_{\mathrm{g}} = \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})$$

2. Contraction energy:

$$e^{(\mathrm{AF})}_{\mathrm{c}} = U(m^2 - n^2) + V \left[2zn^2 - \sum_{\gamma} \left(\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right)\right]$$

3. Entropic energy

$$e_s^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) - \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}}-\tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) + \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \ln\Big(1-f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}}+\tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})$$

4. Chemical potential correction

$$e_n^{(\mathrm{AF})} = \mu n$$

Compose everything

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} \equiv e_{\mathrm{g}}^{(\mathrm{AF})} + e_{\mathrm{c}}^{(\mathrm{AF})} + e_s^{(\mathrm{AF})} + e_n^{(\mathrm{AF})}$$

and we get

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + \ln\Big(1-f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big)\right] + Um^2 - V\sum_{\gamma}\left[\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right] + n\left[\mu + n(2zV-U)\right]$$

As was for the normal phase, the contraction energy and the chemical potential energy corrections build up the RMP $\tilde{\mu}$ of Eq. (1.15),

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \left[\ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + \ln\Big(1-f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big)\right] + Um^2 - V\sum_{\gamma}\left[\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right] + \tilde{\mu} n \tag{1.38}$$

1.6.1 The gap energy

It is useful to see here a property of the above expression. Collecting the terms directly related to the gap function, the following identity holds:

$$Um^2 - V\sum_{\gamma}\left|v^{(\gamma)}\right|^2 = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] \tag{1.39}$$

The proof is simple and immediate: developing the right-hand side,

$$\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] + \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}]$$

Now:

$$\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\} = Um$$

which implies

$$\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] = Um \times \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] = Um^2$$

having we recognized the magnetization from Eq. (1.35). Equivalently,

$$\begin{aligned}\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\} &= -V\sum_{\gamma} v^{(\gamma)}\varphi_{\mathbf{k}}^{(\gamma)*} \\ &= -V\sum_{\gamma} v^{(\gamma)*}\varphi_{\mathbf{k}}^{(\gamma)}\end{aligned}$$

where we used Eq. (1.28) to exchange the conjugation operation in the last passage. This implies:

$$\begin{aligned}\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] &= -V\sum_{\gamma} v^{(\gamma)*}\sum_{\mathbf{k}\in\mathrm{MBZ}}\varphi_{\mathbf{k}}^{(\gamma)}\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}[\beta,\tilde{\mu}] \\ &= -V\sum_{\gamma} v^{(\gamma)*}\times v^{(\gamma)} \\ &= -V\sum_{\gamma}\left|v^{(\gamma)*}\right|^2\end{aligned}$$

where in second passage we recognized $v^{(\gamma)}$ given in Eq. (1.25). Then Eq. (1.39) is verified. The latter is particularly interesting because it shows a non-intuitive result: due to gap complexification, gap amplitude is only increased by $\hat{H}_V$, thus by analogy we would expect the relative free energy contribution to have the same sign as the original gap energy from the real part. This is not the case: to open a gap in the s-wave real sector contributes positively to free energy, while to open a gap in the γ-wave imaginary sector contributes negatively.

An immediate consequence of Eq. (1.39) is that, being the right-hand side evidently positive, for $V > 0$ the total amplitude of the "complexifiers" vector is maximised,

$$\sum_{\gamma}\left|v^{(\gamma)}\right|^2 \leq M \quad \text{being} \quad M^{(\mathrm{AF})} \equiv m^2\frac{U}{V}$$

The above constraint predicts that:

- In the strong coupling limit $U \ll V$, gap complexification is suppressed;
- Correctly, when the physical magnetization closes, so does the imaginary part of the gap

$$\lim_{m\to 0} v^{(\gamma)} = 0$$

  which is coherent with the entire derivation and takes us back to the normal phase.

Eq. (1.39) is also solved by a null magnetization and thus null complexifiers. This can be interpreted by the fact that when the constraint is impossible to solve self-consistently, the gap must close.

[Graphic interpretation: both the aAF and sAF phases have two-syms vs, thus the constraint above tells us that the relative two-elements vector is contained in a circumference.]

1.6.2 Free energy at half-filling

It's of particular interest the half-filling condition, with $\tilde{\mu} = 0$, where PHS discussed in Sec. **??** is preserved in the AF phase. Since, as it can be shown easily,

$$\begin{aligned}\ln\left(1 - \frac{1}{e^x + 1}\right) + \ln\left(1 - \frac{1}{e^{-x} + 1}\right) &= \ln\left[\frac{e^x}{(e^x + 1)^2}\right] \\ &= x - 2\ln(1 + e^x)\end{aligned}$$

Let now $x = \beta\tilde{E}_{\mathbf{k}}$. At low temperature for a band with no nodes, we can approximate $x \to +\infty$, which gives

$$x - 2\ln(1 + e^x) \simeq -x$$

Hence we get the half-filling free energy density substituting ,

$$f_{\mathrm{MFT}}^{(\mathrm{AF})} \simeq -\frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \tilde{E}_{\mathbf{k}} + Um^2 - V \sum_{\gamma} \left[\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2 \right]$$

Starting from the full expression of the free energy density, we recovered the intuitive expression of the energy of a fully (double) occupied band at $-\tilde{E}_{\mathbf{k}}$, with HFPs correction.

[Expand: interesting situation for bands with nodes. At low temperature, where band nodes are present, the function behaves differently because its argument goes to zero. What is the behaviour there?]

1.7 Theoretical constraints on the AF phase HFPs

[To be continued. . .]

1.7.1 Robust C_4 bands shrinking extension

For the normal phase, we saw that hen C_4 symmetry is preserved (Sec. **??**) we can conclude immediately the bands renormalization is actually a shrinking. An identical result holds here: if C_4 symmetry is imposed onto the renormalized bands,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y)$$

with a single self-consistency equation

$$u^{(s^*)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{T}}\mathrm{h}[\beta, \tilde{\mu}]$$

As we did for the normal phase:

1. If $t > 0$ and $u^{(s^*)} \leq 0$
$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \leq 0$$
the equality only holding on the MBZ contour. Looking at the self-consistency equation, all other objects inside the sum are positive-defined. Which gives
$$u^{(s^*)} \leq 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0$$
But $u^{(s^*)} \neq 0$ as long as $t > 0$, as can be seen by the self-consistency equation. We conclude necessarily
$$u^{(s^*)} > 0$$

2. If $t > 0$ and $u^{(s^*)} \geq 2t/V$
$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \geq 0$$
the equality only holding on the MBZ contour. Looking at the self-consistency equation, all other objects inside the sum are positive-defined. Which gives
$$u^{(s^*)} \geq 2t/V \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$
We conclude necessarily
$$u^{(s^*)} < 2t/V$$

The final result,

$$0 < u^{(s^*)} < 2t/V$$

is identical to the constraint obtained for the normal phase. Also in AF phase the bare bands are effectively shrunk by $\hat{H}_V$.

1.7.2 HFPs vanishing in sAF phase

We now present a somewhat convoluted argument to show that, even if raw MFT presents an apparently rich scenario with a lot of HFPs, the problem can be proven to be actually much simpler. Let us start from the sAF phase, described by the five HFPs of Tab. 1.5 and their self-consistency equations of Tabs. 1.3 and 1.4. Let us define the linear functional

$$F[f(\mathbf{k})] \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}[\beta,\tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} f(\mathbf{k})$$

whose linearity properties are trivial,

$$F[af(\mathbf{k})+bg(\mathbf{k})] = aF[f(\mathbf{k})] + bF[g(\mathbf{k})] \tag{1.40}$$

being $a, b \in \mathbb{C}$ and f, g two integrable functions in reciprocal space. Another handy property is the sign-preserving relation for sign-defined functions. Let for instance

$$\forall \mathbf{k} \quad : \quad f(\mathbf{k}) \geq 0 \quad \Longrightarrow \quad F[f(\mathbf{k})] \geq 0 \tag{1.41}$$

This is due to the fact that both $\tilde{\text{Th}}_{\mathbf{k}}$ and $\tilde{E}_{\mathbf{k}}$ are positive-defined. An identical relation with inverted inequality holds for negative-defined function.

Consider now the self-consistency equation for $u^{(s^*)}$: explicitly,

$$\begin{aligned} u^{(s^*)} &= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in \text{MBZ}} (c_x + c_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}_{\mathbf{k}}[\beta,\tilde{\mu}] \\ &= \frac{\left[2t - Vu^{(s^*)}\right]}{L_x L_y} \sum_{\mathbf{k}\in \text{MBZ}} \frac{\tilde{\text{Th}}^{(-)}_{\mathbf{k}}[\beta,\tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x + c_y)^2 - \frac{Vu^{(d)}}{L_x L_y} \sum_{\mathbf{k}\in \text{MBZ}} \frac{\tilde{\text{Th}}^{(-)}_{\mathbf{k}}[\beta,\tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x^2 - c_y^2) \end{aligned}$$

having we used the notation $\cos k_\ell \equiv c_\ell$ and substituted the full bare bands

$$\tilde{\epsilon}_{\mathbf{k}} \equiv -\left[2t - u^{(s^*)}_\sigma V\right](c_x + c_y) + u^{(d)}_\sigma V (c_x - c_y)$$

Evidently, in terms of the functional S, we can rewrite:

$$\begin{aligned} u^{(s^*)} &= \left[2t - Vu^{(s^*)}\right] \times F\left[(c_x+c_y)^2\right] - Vu^{(d)} \times F\left[c_x^2 - c_y^2\right] \\ &= \left[2t - Vu^{(s^*)}\right] \times \left\{F\left[c_x^2\right] + 2F\left[c_x c_y\right] + F\left[c_y^2\right]\right\} - Vu^{(d)} \times \left\{F\left[c_x^2\right] - F\left[c_y^2\right]\right\} \end{aligned}$$

having we used the linearity property of Eq. (1.40) in last passage.

Let now:

$$A \equiv F\left[c_x^2\right] \qquad B \equiv 2F\left[c_x c_y\right] \qquad C \equiv F\left[c_y^2\right]$$

which allows us to rewrite the self-consistency equation as:

$$u^{(s^*)} = \left[2t - Vu^{(s^*)}\right](A+B+C) - Vu^{(d)}(A-C)$$

Repeating an identical procedure on the three self-consistency equations for $u^{(d)}$, $v^{(s^*)}$ and $v^{(d)}$, we get the full system:

$$u^{(s^*)} = \left[2t - Vu^{(s^*)}\right](A+B+C) - Vu^{(d)}(A-C) \tag{1.42}$$

$$u^{(d)} = \left[2t - Vu^{(s^*)}\right](A-C) - Vu^{(d)}(A-B+C) \tag{1.43}$$

$$v^{(s^*)} = -Vv^{(s^*)}(A+B+C) - Vv^{(d)}(A-C) \tag{1.44}$$

$$v^{(d)} = -Vv^{(s^*)}(A-C) - Vv^{(d)}(A-B+C) \tag{1.45}$$

These equations define a four-fold nonlinear map, whose parameters A, B, C depend on the HFPs themselves, being the functional F dependent on the HFPs. Because of this last point, the four equations are coupled to each other and the map does not decouple in two independent sub-maps.

Aim of this section is to prove that no self-consistent solution exists with non-zero $u^{(d)}$, $v^{(s^*)}$ and $v^{(d)}$. Being the derivation convoluted and abstract enough, let us proceed by enumerating the relevant facts.

1. Let us start by applying a conformal transform on the map: define the new "sum" and "difference" variables

$$S_u \equiv u^{(s^*)} + u^{(d)} \qquad D_u \equiv u^{(s^*)} - u^{(d)} \qquad S_v \equiv v^{(s^*)} + v^{(d)} \qquad D_v \equiv v^{(s^*)} - v^{(d)}$$

whose matrix representation is

$$\begin{bmatrix} S_u \\ D_u \\ S_v \\ D_v \end{bmatrix} \equiv \begin{bmatrix} 1 & 1 & & \\ 1 & -1 & & \\ & & 1 & 1 \\ & & 1 & -1 \end{bmatrix} \begin{bmatrix} u^{(s^*)} \\ u^{(d)} \\ v^{(s^*)} \\ v^{(d)} \end{bmatrix}$$

The map transform is obtained by summing and subtracting two couples of equations: Eq. (1.42) and (1.43) (the first two) as well as Eq. (1.44) and (1.45) (the second two). By doing this we get the simplified system:

$$(1 + 2AV)S_u + BVD_u = 2t(2A + B) \tag{1.46}$$
$$(1 + 2CV)D_u + BVS_u = 2t(B + 2C) \tag{1.47}$$
$$(1 + 2AV)S_v + BVD_v = 0 \tag{1.48}$$
$$(1 + 2CV)D_v + BVS_v = 0 \tag{1.49}$$

2. Let $S_v = 0$. Then $D_v = 0$. From Eq. (1.49), using $C = F\left[c_y^2\right] \geq 0$ implied by Eq. (1.41):

$$\underbrace{(1 + 2CV)}_{>0} D_v = 0 \qquad \Longrightarrow \qquad D_v = 0$$

Identically, let $D_v = 0$. Then applying the same consideration on Eq. (1.48), $S_v = 0$. Either both are non-zero or both are zero.

3. Let $S_v \neq 0$ and $D_v \neq 0$. Then $B \neq 0$. To prove this, consider for example Eq. (1.48): we are working at $V > 0$, thus

$$B = -\frac{1 + 2AV}{V} \frac{S_v}{D_v}$$

which is non-zero because $A = F\left[c_x^2\right] \geq 0$, the latter following from Eq. (1.41). We could have used identically Eq. (1.49).

4. Let $S_v \neq 0$ and $D_v \neq 0$. Hence the following relation holds:

$$(BV)^2 = (1 + 2CV)(1 + 2AV) \tag{1.50}$$

To prove the latter, it is sufficient to multiply Eq. (1.48) by $BV \neq 0$ (see point above),

$$BV(1 + 2AV)S_v + (BV)^2 D_v = 0$$

and substitute from Eq. (1.49) $BVS_v = -(1 + 2CV)D_v$,

$$\left[-(1 + 2CV)(1 + 2AV) + (BV)^2\right] D_v = 0$$

Since $D_v \neq 0$ by hypothesis, Eq. (1.50) is implied.

5. Let $S_v \neq 0$ and $D_v \neq 0$. Hence the following relation holds:

$$B + 2C = V(B^2 - 4AC) \tag{1.51}$$

To prove this: multiply Eq. (1.46) by BV on both sides,

$$BV(1 + 2AV)S_u + (BV)^2 D_u = 2tBV(2A + B)$$

and substitute from Eq. (1.47) $BVS_v = -(1+2CV)D_v + 2t(B+2C)$,

$$\begin{aligned}
-(1+2CV)(1+2AV)D_v + +2t(B+2C)(1+2AV) + (BV)^2 D_u &= 2tBV(2A+B) \\
2t(B+2C)(1+2AV) &= 2tBV(2A+B) \\
B + 2ABV + 2C + 4ACV &= 2ABV + B^2 V
\end{aligned}$$

Passing from first to second line we used Eq. (1.50), and in the passage below $t > 0$. Last line implies Eq. (1.51).

6. Let $S_v \neq 0$ and $D_v \neq 0$. Then $B^2 \neq 4AC$. Indeed, let $B^2 = 4AC$. Then from Eq. (1.50),

$$(BV)^2 = 1 + 2V(A+C) + 4ACV^2 \qquad \Longrightarrow \qquad 1 + 2V(A+C) = 0$$

which is obviously absurd, being $A \geq 0$, $C \geq 0$ from Eq. (1.41). A consequence of this implication on Eq. (1.51) is that, at a given V, the map parameters should satisfy

$$V = \frac{B+2C}{B^2 - 4AC} \tag{1.52}$$

7. Let $S_v \neq 0$ and $D_v \neq 0$. Then $A = C$. To prove this, consider the parabolic solutions of Eq. (1.51),

$$(B^2 - 4AC)V^2 - 2V(A+C) - 1 = 0 \implies V_{1,2} = \frac{(A+C) \pm \sqrt{(A+C)^2 + (B^2 - 4AC)}}{B^2 - 4AC}$$

To identify a fixed point of the map, at least one of those points should satisfy Eq. (1.52). Then it must be

$$\begin{aligned}
A + C \pm \sqrt{(A+C)^2 + (B^2 - 4AC)} &= B + 2C \\
(A+C)^2 + (B^2 - 4AC) &= \left[(B+2C) - (A+C)\right]^2 \\
B^2 - 4AC &= (B+2C)^2 - 2(B+2C)(A+C) \\
B^2 - 4AC &= B^2 + 4C^2 + 4BC - 2AB - 2BC - 4AC - 4C^2
\end{aligned}$$

Simplifying all recurring terms, we get

$$2B(A-C) = 0$$

From previous points we know $B \neq 0$. Then $A = C$.

8. It's impossible to have $S_v \neq 0$ and $D_v \neq 0$. To prove this, assume the contrary and use the point above to reduce Eq. (1.52) to

$$A = C \qquad \Longrightarrow \qquad V = \frac{B+2C}{B^2 - 4C^2} = \frac{1}{B-2C}$$

Inserting the latter in Eq. (1.49),

$$\begin{aligned}
\left(1 + \frac{2C}{B-2C}\right) D_v + \frac{B}{B-2C} S_v &= 0 \\
\frac{B}{B-2C} D_v + \frac{B}{B-2C} S_v &= 0
\end{aligned}$$

which implies

$$D_v + S_v = 0 \qquad \Longrightarrow \qquad v^{(s^*)} = 0$$

Recalling from Tab. 1.4 the explicit self-consistency equation for $v^{(d)}$ and setting $v^{(s^*)} = 0$,

$$v^{(d)} = -V v^{(d)} F\left[(c_x - c_y)^2\right]$$

and using the positivity rule of Eq. (1.41) we get

$$V > 0 \quad \text{and} \quad F\left[(c_x - c_y)^2\right] \geq 0 \qquad \Longrightarrow \qquad v^{(d)} = 0$$

because no finite positive number can equal any finite negative number, and viceversa. Thus

$$S_v \neq 0 \quad \text{and} \quad D_v \neq 0 \qquad \Longrightarrow \qquad v^{(s^*)} = v^{(d)} = 0 \qquad \Longrightarrow \qquad S_v = D_v = 0$$

Absurd. We explored any possibility for S_v, D_v: no stable solution with finite values of the HFPs $v^{(s^*)}$, $v^{(d)}$ exists, thus both must be zero and can be eliminated from the algorithm.

[To be continued...]

1.7.3 HFPs vanishing in aAF phase

Bibliography

[1] James F. Annett. “Symmetry of the order parameter for high-temperature superconductivity”. In: *Advances in Physics* 39.2 (Apr. 1990). Publisher: Taylor & Francis _eprint: https://doi.org/10.1080/00018739000101481, pp. 83–126. ISSN: 0001-8732. DOI: 10.1080/00018739000101481. URL: https://doi.org/10.1080/00018739000101481 (visited on 01/06/2026).

[2] Daniel P. Arovas et al. “The Hubbard Model”. In: *Annual Review of Condensed Matter Physics* 13.Volume 13, 2022 (2022), pp. 239–274. ISSN: 1947-5462. DOI: https://doi.org/10.1146/annurev-conmatphys-031620-102024. URL: https://www.annualreviews.org/content/journals/10.1146/annurev-conmatphys-031620-102024.

[3] Anna E. Böhmer et al. “Nematicity and nematic fluctuations in iron-based superconductors”. en. In: *Nature Physics* 18.12 (Dec. 2022), pp. 1412–1419. ISSN: 1745-2481. DOI: 10.1038/s41567-022-01833-3. URL: https://www.nature.com/articles/s41567-022-01833-3.

[4] Zhangkai Cao et al. “Dominant p-wave pairing induced by nearest-neighbor attraction in the square-lattice extended Hubbard model”. In: *Phys. Rev. B* 111 (2 Jan. 2025), p. 024509. DOI: 10.1103/PhysRevB.111.024509. URL: https://link.aps.org/doi/10.1103/PhysRevB.111.024509.

[5] Piers Coleman. *Introduction to Many-Body Physics*. Cambridge University Press, 2015.

[6] Michele Fabrizio. *A Course in Quantum Many-Body Theory*. Springer, 2022.

[7] Gabriele Giuliani and Giovanni Vignale. *Quantum Theory of the Electron Liquid*. Cambridge University Press, 2005.

[8] Giuseppe Grosso and Giuseppe Pastori Parravicini. *Solid State Physics*. Second Edition. Academic Press, 2014.

[9] J. E. Hirsch. “Two-dimensional Hubbard model: Numerical simulation study”. In: *Phys. Rev. B* 31 (7 Apr. 1985), pp. 4403–4419. DOI: 10.1103/PhysRevB.31.4403. URL: https://link.aps.org/doi/10.1103/PhysRevB.31.4403.

[10] Joel Hutchinson and Frank Marsiglio. “Mixed temperature-dependent order parameters in the extended Hubbard model”. en. In: *Journal of Physics: Condensed Matter* 33.6 (Feb. 2021), p. 065603. ISSN: 0953-8984, 1361-648X. DOI: 10.1088/1361-648X/abc801. URL: https://iopscience.iop.org/article/10.1088/1361-648X/abc801 (visited on 01/06/2026).

[11] R. Micnas, J. Ranninger, and S. Robaszkiewicz. “Superconductivity in narrow-band systems with local nonretarded attractive interactions”. In: *Reviews of Modern Physics* 62.1 (Jan. 1990). Publisher: American Physical Society, pp. 113–171. DOI: 10.1103/RevModPhys.62.113. URL: https://link.aps.org/doi/10.1103/RevModPhys.62.113 (visited on 01/06/2026).

[12] Robin Scholle et al. “Comprehensive mean-field analysis of magnetic and charge orders in the two-dimensional Hubbard model”. In: *Phys. Rev. B* 108 (3 July 2023), p. 035139. DOI: 10.1103/PhysRevB.108.035139. URL: https://link.aps.org/doi/10.1103/PhysRevB.108.035139.

[13] P. Senarath Yapa et al. “Mixed-symmetry superconductivity and the energy gap”. en. In: *Physica C: Superconductivity and its Applications* 633 (June 2025), p. 1354719. ISSN: 09214534. DOI: 10.1016/j.physc.2025.1354719. URL: https://linkinghub.elsevier.com/retrieve/pii/S0921453425000723 (visited on 01/06/2026).

[14] Manfred Sigrist and Kazuo Ueda. “Phenomenological theory of unconventional superconductivity”. In: *Reviews of Modern Physics* 63.2 (Apr. 1991). Publisher: American Physical Society, pp. 239–311. DOI: 10.1103/RevModPhys.63.239. URL: https://link.aps.org/doi/10.1103/RevModPhys.63.239 (visited on 01/07/2026).

[15] Avinash Singh and Zlatko Tešanović. “Collective excitations in a doped antiferromagnet”. en. In: *Physical Review B* 41.1 (Jan. 1990), pp. 614–631. ISSN: 0163-1829, 1095-3795. DOI: 10.1103/PhysRevB.41.614. URL: https://link.aps.org/doi/10.1103/PhysRevB.41.614 (visited on 12/12/2025).

[16] Ashvin Vishwanath. “Dirac Nodes and Quantized Thermal Hall Effect in the Mixed State of d-wave Superconductors”. In: *Physical Review B* 66.6 (Aug. 2002). arXiv:cond-mat/0104213, p. 064504. ISSN: 0163-1829, 1095-3795. DOI: 10.1103/PhysRevB.66.064504. URL: http://arxiv.org/abs/cond-mat/0104213 (visited on 01/05/2026).

[17] Alexander Wietek et al. “Stripes, Antiferromagnetism, and the Pseudogap in the Doped Hubbard Model at Finite Temperature”. en. In: *Physical Review X* 11.3 (July 2021), p. 031007. ISSN: 2160-3308. DOI: 10.1103/PhysRevX.11.031007. URL: https://link.aps.org/doi/10.1103/PhysRevX.11.031007 (visited on 12/12/2025).

[18] C. Zhou and H. J. Schulz. “Density of states and tunneling spectra in two-dimensional d-wave superconductors”. In: *Physical Review B* 45.13 (Apr. 1992). Publisher: American Physical Society, pp. 7397–7401. DOI: 10.1103/PhysRevB.45.7397. URL: https://link.aps.org/doi/10.1103/PhysRevB.45.7397 (visited on 01/06/2026).